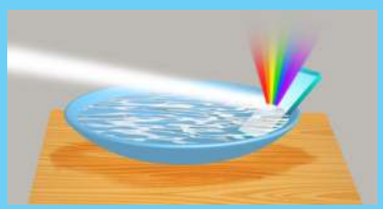
Difference between the images formed in Pinhole camera and Plane mirror	
Images formed by Pin hole camera	**Images formed in Plane mirror**
The image is real	The image is virtual
The image may not be equal to the size of the object	The image is equal to the size of the object
The image is inverted	The image is erect

ACTIVITY 10

Let's make a rainbow

You must surely have seen a rainbow in the sky. Why don't you try making one at your class room?



Place a flat but deep pan. Place this pan where there is direct sunlight. Place a plain mirror in the pan as shown in the diagram, so that you see sunlight reflected on you ceiling or on a white wall. Next slowly pour water in to the pan. At particular level of water, you will get a beautiful rainbow colours on the wall. If the colours are not clear adjust the position of the mirror to bring it into focus. This arrangement of colours in sunlight is called spectrum.

DO YOU KNOW?

Why danger lights in vehicles are red in colour?



1. Red colour is scattered the least by air molecules.
2. Red colour has the highest wavelength of all the other colours. So red colour is able to travel the longest distance through air, fog.

When white light is passed through a prism as shown in the figure, the colours of the rainbow emerge from the prism. This phenomenon is called **dispersion of light**, and the band of colour is called as **spectrum**.

Newton Disc:

Newton suggested a process of mixing different colours to make white colour by setting an arrangement as shown figure below. Newton Disc is a card board disc with seven equal sectors coloured red, yellow, orange, green, blue, indigo and violet. When the disc turned quickly, the retina receives the sensation of the spectrum simultaneously and disc appears white. Using this disc, one can explain that white is a combination of VIBGYOR.



VII STD Science Term-3 EM unit 1.indd 17 9/30/2023 12:22:46 PM